Rithvik Vootukur

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Education

Texas A&M University College Station, TX
Bachelor of Science in Aerospace Engineering | Engineering Honors Program | GPA: 3.70/4.00 *Expected May 2027*

- **Relevant Coursework:** Aerodynamics, Aerospace Structures, Spacecraft Design, Propulsion, Heat Transfer, Numerical Methods, Machine Learning

Research Interests

Laser beam propagation through turbulence; turbulence-scale filtering; beam wander and scintillation; aerospace numerical modeling; spacecraft systems.

Research Experience

Research Assistant at Turbulence and Advanced Computations Laboratory Jun 2026 – Present
TACL, Texas A&M University | Advisor: Prof. Diego Donzis *College Station, TX*

- Analyze RIPTIDE simulation outputs to study how filtering turbulence scales just above the Kolmogorov microscale affects laser beam wander and scintillation.
- Verified and visualized a 4,218-case RIPTIDE sweep (111 focal-length and propagation-distance pairs, each with 37 filter scales from 0.1 mm to 3 m) in Python and Matplotlib on Linux.
- Present research plots and interpretations that guide which conditions and turbulence scales to study next.

AI-Assisted Aircraft Geometry Generation | *Professor-guided research, Texas A&M University* 2025 – 2026

- Developed structured prompting workflows for aircraft-geometry point sets exported as CSV data; prototyped checks to screen generated geometry before aerodynamic or stability validation.

Hypersonic Machine Learning Exploration | *Professor-guided research, Texas A&M University* 2026

- Applied exploratory logistic regression to examine associations between rarefaction-related variables and reentry ionization behavior.

Engineering Leadership

Team Lead at TAMU-SPIRIT-2 Aug 2026 – Present
Senior Design Capstone (AERO 402), Texas A&M University *College Station, TX*

- Lead a 16-person team in a Texas A&M and Aegis Aerospace program preparing four experiments for an external International Space Station (ISS) platform based on the MISSE architecture.
- Edited and presented the System Definition Review (SDR), covering mission goals, requirements traced to goals with verification methods, concept of operations, and the plan to Preliminary Design Review (PDR).
- Contribute hands-on to the structures and electronics subsystems: run Abaqus finite element analysis (FEA) for structural analysis; conduct battery research and trade studies for thermal and power planning.

Project Manager and Team Lead, NASA Gateways to Blue Skies Competition 2023 – 2024
Texas A&M University *College Station, TX*

- Led a six-member multidisciplinary team to submit a 14-page technical proposal for a conceptual spherical search-and-rescue unmanned aerial vehicle (UAV); managed subsystem work and the project schedule.
- Originated the roll-cage concept and developed the SolidWorks CAD model, geometry sizing, and weight estimates; supported materials selection with Pugh-chart trade studies.
- Wrote technical documentation on nanomaterials, produced Blender renders and animation, and delivered an approximately 20-minute technical presentation with Q&A.

Presentations

TAMU-SPIRIT-2 System Definition Review (SDR) | *Presenter and editor, Texas A&M University* Oct 2026

- Presented the program mission, system goals, requirements, architecture, and plan to Preliminary Design Review (PDR).

NASA Gateways to Blue Skies Competition Proposal | *Presenter, Texas A&M University* 2023 – 2024

- Delivered an approximately 20-minute technical presentation with Q&A on a conceptual spherical search-and-rescue UAV.

Projects

AS 35 Mi Sailplane Wing Design | *Aerodynamic design project, SolidWorks, XFOIL*

- Built a parametric SolidWorks model of the wing and surrounding aircraft geometry to support configuration changes during design iteration.
- Compared 10 glider airfoil candidates using aerodynamic performance metrics and documented a trade study to support design selection.

RS-25 Nozzle and Compressible Flow Solver | *Numerical analysis project, Python*

- Developed a Python solver for one-dimensional supersonic nozzle flow using the area-Mach relation and bracketing root finding to compute Mach number, pressure, temperature, density, velocity, and nozzle area from throat to exit.
- Vacuum thrust came within 5.1% of public RS-25 data; traced larger specific impulse and mass-flow gaps to the assumed gas constant in the simplified one-dimensional model.

Heat Transfer Numerical Solvers | *Numerical methods projects*

- Implemented second- and fourth-order finite difference and first- and second-order finite element formulations for one-dimensional steady conduction with convection and material interfaces.
- Assembled global systems, applied Dirichlet boundary conditions with a penalty method, and confirmed the expected 2nd- and 4th-order convergence with Richardson extrapolation.

Bearing Fault Classification | *Independent extension of course project, Python, PyTorch*

- Processed 40 Case Western Reserve University (CWRU) vibration recordings; normalized signals and applied Fast Fourier Transform (FFT) preprocessing to normal, inner-race, outer-race, and ball-fault cases.
- Trained scikit-learn logistic regression and a PyTorch neural network on FFT spectra, reaching 98.5% accuracy (macro F1 0.99) on held-out motor loads, up from 72.5% for a hand-crafted-feature baseline.

Professional Memberships

American Institute of Aeronautics and Astronautics (AIAA); American Society of Mechanical Engineers (ASME)

Technical Skills

Programming: Python, C++, Fortran, MATLAB
Engineering Design: Computer-aided design (CAD), SolidWorks, Abaqus, finite element analysis (FEA), XFOIL
Machine Learning: scikit-learn, PyTorch; supervised learning, logistic regression, neural networks
Numerical Methods: Finite element method (FEM), finite differences, root finding, Richardson extrapolation
Scientific Data: NumPy, SciPy, HDF5; data preprocessing, signal processing, Fast Fourier Transform (FFT)
Visualization and Computing: Matplotlib, Blender, Linux, Git, SVN version control